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From ecology to outbreaks: Integrating knowledge for effective vector management in Europe

Mosquito population mixing shapes Rift Valley Fever transmission across a wildlife–livestock interface

Merline Feero[1], Tino Pori[1], Peter Thompson[2], Bree Beechler[3], Anna Jolles[3], Morgan Bar[1], Erin Gorsich[1]

[1]University of Warwick, UK [2]University of Pretoria, South Africa [3]Oregon State University, USA

Merline.Feero@warwick.ac.uk

Introduction

Rift Valley Fever (RVF) is a mosquito-borne zoonosis of domestic and wild ruminants. Between epidemics in domestic ruminants, the virus continues to circulate in African buffalo in Kruger National Park (KNP),[1,2] and exposure has been documented in numerous wildlife species.[3] Recent research has shown that RVF can circulate at low levels, undetected in cattle populations.[4]

It is currently unknown how RVF circulates at wildlife–livestock interfaces.

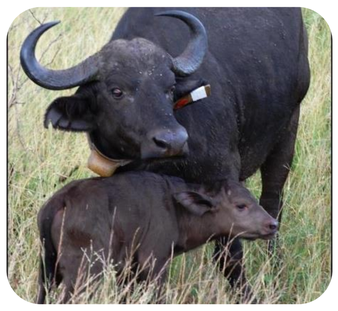


Figure 1. *African buffalo cow and calf, Kruger National Park.*

Objectives

- Measure RVF seroprevalence and incidence simultaneously in buffalo and cattle
- Characterise mosquito blood-feeding on wildlife and livestock across the park boundary
- Build a mechanistic model of RVF across the wildlife–livestock interface
- Investigate how mosquito mixing contributes to RVF circulation in this system

Methodology

Mosquito and host sampling

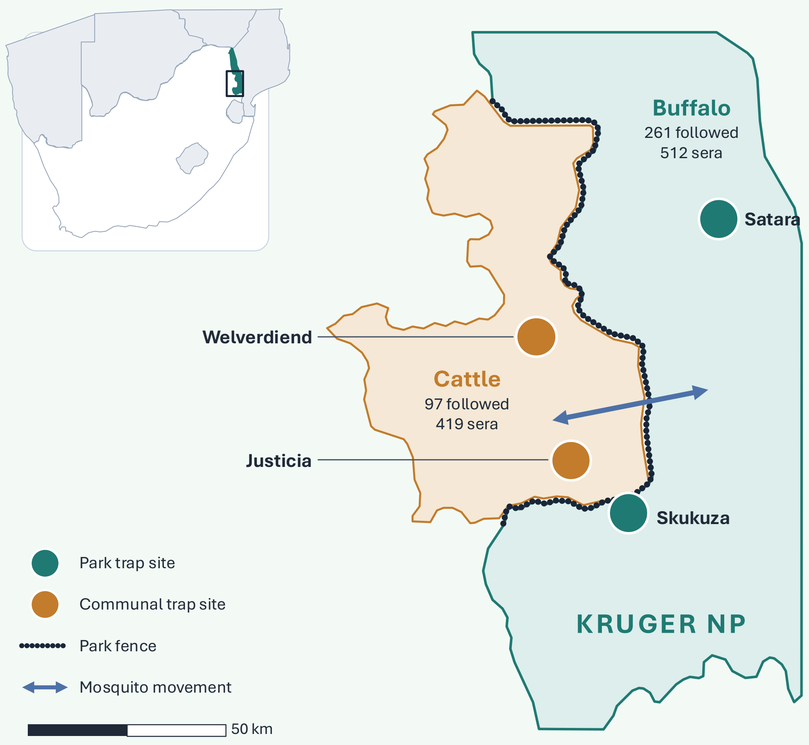






Figure 2. *Study sites of mosquito trapping and host serology on both sides of the park fence. Example of mosquito trap types used (left: BG Sentinel; right: net trap).*

Mechanistic model

Buffalo and cattle (SIR); *Aedes* and *Culex* (SEI) mosquitoes are represented. Other inputs include host preference from the blood meals; temperature-driven mosquito life history[5]; infected *Aedes* eggs carry virus through the dry season (transovarial transmission).[1,6] Mosquitoes cross the fence to feed and return home; the share of bites taken across is represented by ω.

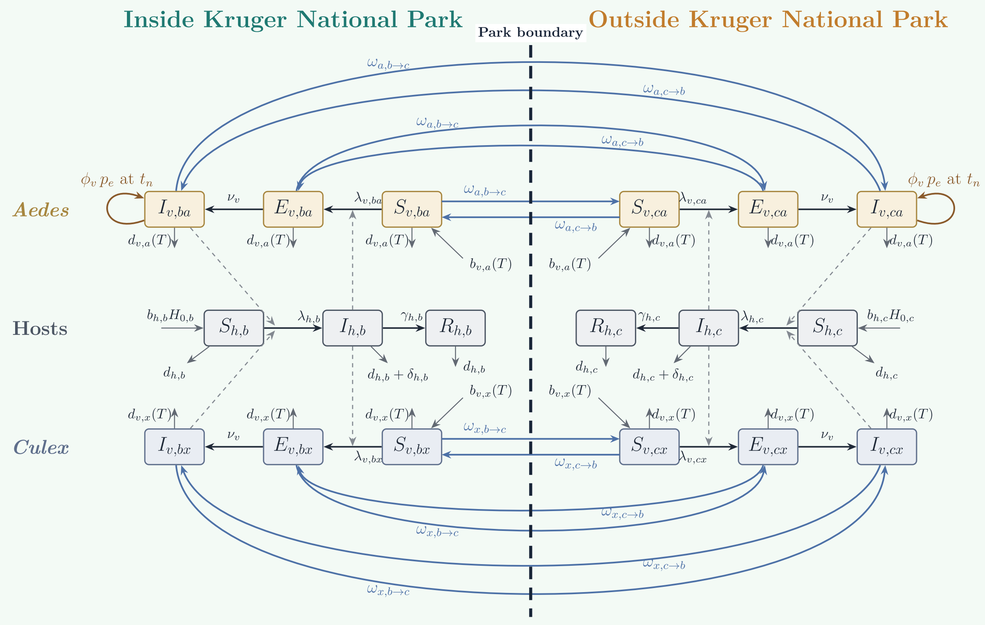
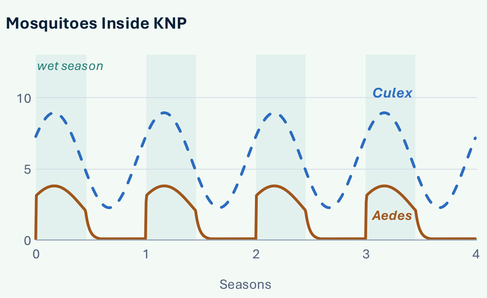


Figure 3. *Model structure, either side of the park boundary.*

- *Progression through states*
- *Births and deaths*
- *Infection*
- *Mosquitoes crossing the fence to feed (ω)*
- *Annual transovarial transmission (*Aedes *only)*



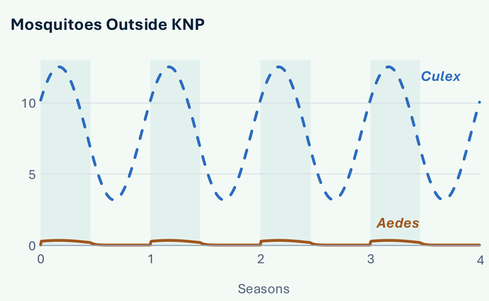


Figure 4. *Adult mosquito dynamics per host over several seasons.* Aedes *(brown) emerge from non-permanent water sources at the beginning of the wet season (shaded) and die back afterwards, persisting at a low dry-season population (matched to field catches[7]);* Culex *(blue) follow temperature seasonality and carrying capacity due to breeding in permanent water sources.*

Results

A RVF prevalence and incidence in buffalo, in the absence of an outbreak in cattle

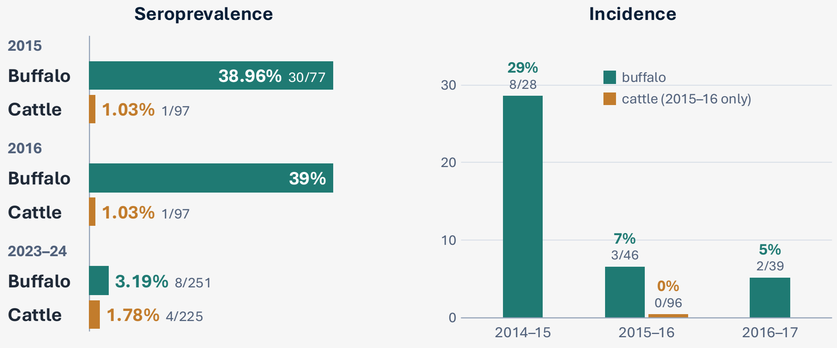


Figure 5A. *Left: RVF seroprevalence (positive/tested) in buffalo and cattle. Right: yearly incidence in the longitudinal herds as the share of seronegative animals that seroconverted. Unpublished data, T. Pori.*

B Blood meal analysis data quantifies mosquito feeding across the park fence

We collected **292** blood-fed mosquitoes from the subfamily Culicinae, representing **47 mosquito species** feeding on **29 vertebrate blood meal sources**, including cattle, chicken, impala, and kudu.

Trap location type	Wildlife blood meals	Livestock blood meals
Park	16	1
Communal land	4	10

Scan for full host–vector feeding network

Figure 5B. *Shaded cells represent blood meals taken on the other side of the park fence. Unpublished data, T. Pori.*

C Models show that while buffalo sustain RVF, cross-boundary feeding is a bridge to cattle infection

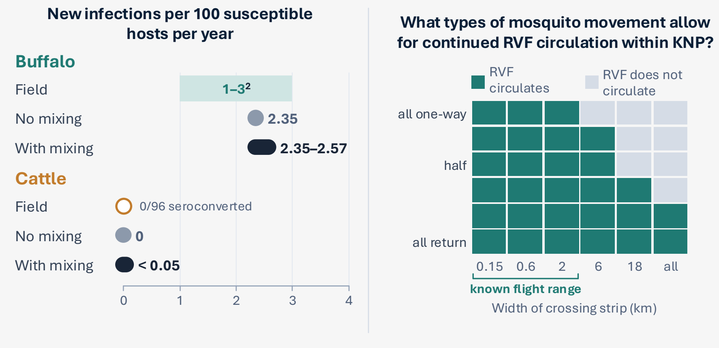


Figure 5C. *Left: modelled incidence with and without mosquito mixing. Field values: free-ranging buffalo[2]; 0/96 cattle. Right: KNP can maintain RVF unless mosquitoes move one-way beyond known flight ranges. Model results are preliminary.*

Conclusion

At the Kruger National Park interface, RVF can circulate in buffalo independently of cattle. Cattle exposure depends on mosquitoes feeding across the park boundary, and, at the field-measured mixing rates, stays below detection.



Acknowledgements

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1 Manore & Beechler 2015 Transbound Emerg Dis 62:13 · **2** Beechler et al. 2015 Transbound Emerg Dis 62:24 · **3** LaBeaud et al. 2011 Am J Trop Med Hyg 84:641 · **4** van den Bergh et al. 2019 PLoS Negl Trop Dis 13:e0007296 · **5** Mordecai et al. 2019 Ecol Lett 22:1690 · **6** Chitnis et al. 2013 J Biol Dyn 7:11 · **7** Guarido et al. 2021 Parasit Vectors 14:339